

Lecture 1 (Introduction)

procedures for the definition of techniques for the design of Electrical drives
with usage of MATLAB/SIMULINK for simulation

EXAM ORAL exam (eng/ita) + report about an exercise activity
↳ 2 questions about dynamic model of some electrical machine

(organized by appointment, 10 days before) ⇒ also if during official session send anyway an email to program the examination

Example OLDOGRAPHIC Ventilation

using an electrical drives subjected to disturbances → to reject the disturbance you have to

carefully on the BANDWIDTH ← design the CONTROL LOOP appropriately for the type of disturbance
feedback given on the system?

The control of the system is given by a position sensor (by speed integration? NO)

a cheaper solution to measure the position, instead of a complex sensor you can use another solution → absolute position of the system by HALL EFFECT sensor

that give the permanent magnet position correctly. → using the permanent magnet on the rotor to obtain a correct position measurements

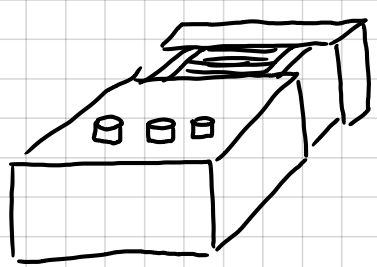
(otherwise it can be used an ENCODER)

disturbing the permanent magnet the system acts badly → if using also an encoder (with good step precision you can increase control performances on the DC machine and change led configuration)

↓ this is a system with a POWER supply directly connected, with constant speed of Rotor, NO power control



Example AC power supply

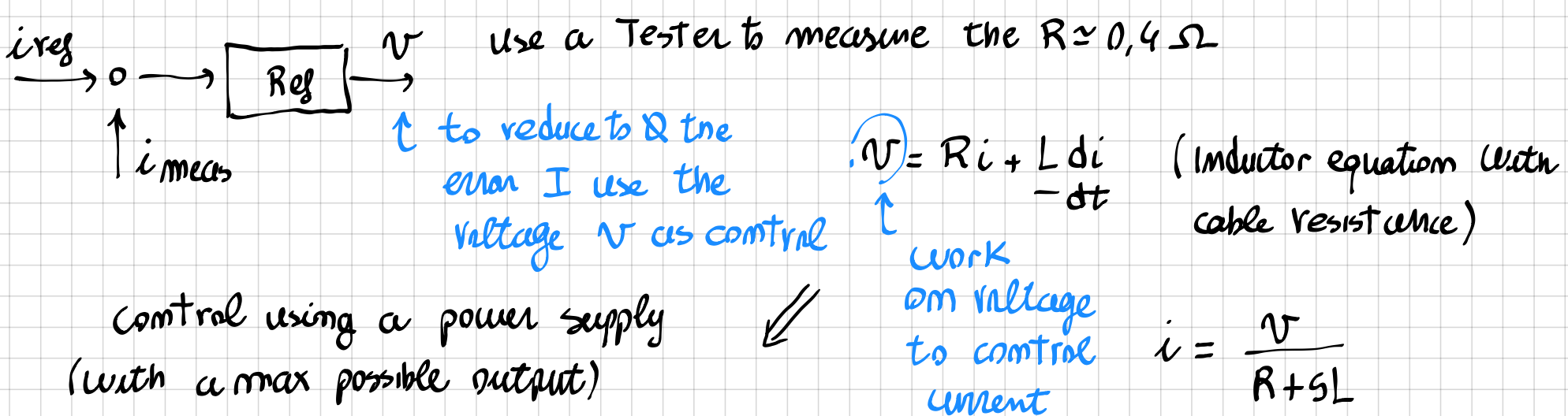


give you info about nominal current → the value of current that allows the device to have a long life-time

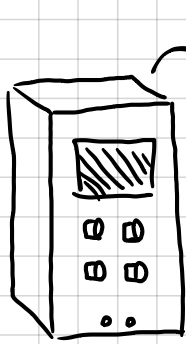
if you know the system is cheaper an OPEN LOOP simple approach without sensors

↳ but with OPEN LOOP there is NO feedback, risky if you need precision

Open Loop if I want to power supply it correctly I may use a tester to measure resistance (if closed loop I need to measure the current feedback)

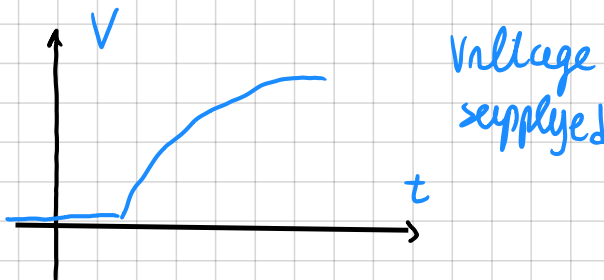


control using a power supply (with a max possible output)



connected to an oscilloscope

OPEN LOOP approach



Looking this you can infer the time constant $\tau = L/R$ of the system

(the open loop can help the closed loop) ← feed forward using syst knowledge

IF you want to reduce the time constant τ , you increase the voltage

how to reduce the transient time... by an

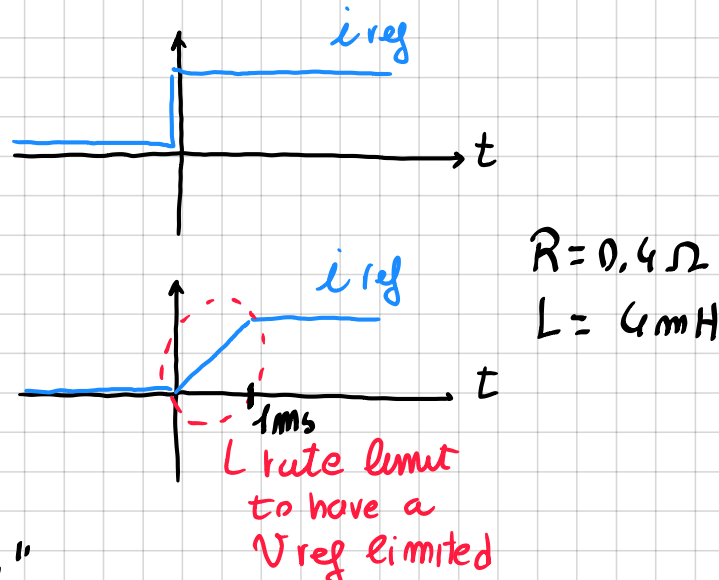
increase of R with an external Resistance ⇒ here the problem is caused by the open loop approach

from

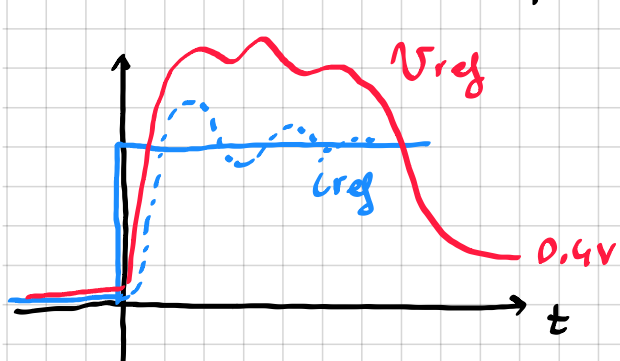
$$V = Ri + L \frac{di}{dt} \Rightarrow V_{reg} = Ri_{reg} + L \frac{di_{reg}}{dt}$$

performance limitation due to open loop control

$$V_{reg}(0) = 4V + 0.4 = 4.4V$$



while on Closed Loop



PI control

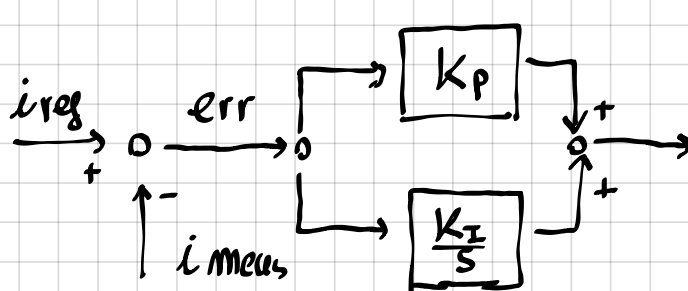
(not D because "derivative"

is hard to obtain) Proportional + Integrational control

design of controller made on

low error → if high error

bad design → SATURATION



Using an ANTI-WIND UP (to solve saturation issue)

manage the saturation on the system

Similar approach on MACHINE, with also torque acting on the system

You can use { open loop close loop } help close loop with parallel feed forward approach